Repeat purchase prediction project checkpoint

Group 1 | Devano Jonathan | 3036029698 | 5 November 2026

Status. I am Jonathan, the text-feature owner for Group 1. The repaired notebook runs directly from the checked-in data archive, completes all 27 code cells and passes 19 contract tests. I reviewed the methodology, replaced the weak sparse TF-IDF comparison with LSA-TF-IDF, checked the MiniLM benchmark and error outputs, and updated WORK_SPLIT.md to reflect the completed work.

Project progress

Dataset and design. The project uses the UCI Online Retail dataset (ID 352), covering 541,909 invoice lines from 1 December 2010 to 9 December 2011. The v2 pipeline keeps 391,153 merchandise product lines and separates 1,539 postage, carriage, adjustment, fee and voucher lines. Month-end snapshots from March to October use a 90-day feature window and a 30-day target. March to June trains, July selects, and August to October tests chronologically.

Data audit item	Value	Data audit item	Value
Raw invoice lines	541,909	Missing customer IDs	135,080
Exact duplicate rows	5,268	Missing descriptions	1,454
Merchandise product lines	391,153	Merchandise invoices	18,402
Customer-snapshot rows	16,123	Modelled monthly cohorts	8

Data decision. The one-year dataset remains the main source because every feature and target window is fully covered. Online Retail II adds no useful coverage for these dates: its 2010-2011 sheet is the same target-year data plus one postage line, and its two sheets repeat 22,523 rows from 1-9 December 2010. Exact duplicate removal changes 0.27 percent of valid revenue and none of the eight monthly target labels.

Evidence from the current pipeline

Finding. I checked the text representations before interpreting their model gains. LSA-TF-IDF raises the best cosine silhouette from 0.015 for raw sparse KMeans to 0.145 at K=20, and its largest cluster share falls from about 68 percent to 25 percent. MiniLM reaches 0.075 at K=20. A matched K=16 comparison shows TF-IDF at 0.123 and MiniLM at 0.067.

Model or baseline	July PR-AUC	Aug-Oct PR-AUC
Majority baseline	0.317	0.406
Recency rule	0.407	0.418
Logistic RFM	0.598	0.625
Boosting, behaviour	0.608	0.633
Boosting + LSA-TFIDF	0.615	0.634
Boosting + MiniLM	0.612	0.634

Interpretation. Boosting plus LSA-TF-IDF is the July-selected model, with validation PR-AUC 0.615, final pooled PR-AUC 0.634 and lift at 20 percent of 1.77. The final test does not replace July selection. Paired intervals place the leading learned models within overlapping ranges, so the report treats them as indistinguishable instead of choosing a winner from a third-decimal difference.

Individual focus and evidence

Devano Jonathan. I own stages 8-9 and the interpretation of October errors. I reviewed both best-K and matched-K cluster results, checked representative product descriptions and compared false positives with false negatives. In the October snapshot, false positives have median recency 22.46 days, while false negatives have median recency 36.51 days. The active false-positive group explains why a short recency score does not separate every returning customer.

Concrete output. My evidence is in product_cluster_examples.csv, cluster_quality.csv, cluster_matched_k_quality.csv, error_analysis_summary.csv and paired_model_differences.csv. I also reviewed the executed notebook, 19-test contract suite and rendered report. My remaining personal tasks are to confirm the descriptive cluster labels and review the final checkpoint wording.

Group contributions

Member	Work completed or in progress	Evidence or output	Approx. hours	Next checkpoint task
Stacia Benita	Shared EDA, product taxonomy, revenue audit and duplicate check are complete; personal review is still required.	EDA figures, cleaning audit and revenue audit.	18	Review the data evidence and independently rerun her stage.
Devano Jonathan	Completed the methodology and text-feature review, LSA comparison, matched K=16 check, error interpretation and checkpoint update.	Cluster-quality tables, cluster examples, error summary and 19 passing tests.	20	Accept the cluster descriptions and review the final wording.
Surjadjajadi Clara Natalie	Shared snapshot, model and uncertainty outputs are complete; final pipeline review and report merge remain.	Validation, monthly results and paired interval files.	19	Verify the pipeline section and review the merged report.

Approximate hours: Stacia Benita 18, Devano Jonathan 20, Clara Natalie Surjadjajadi 19.

Next steps and blockers

Target date	Owner	Task
6 Nov 2026	All	Complete individual checkpoint evidence: data sensitivity, cluster and error review, and verified pipeline outputs.
7 Nov 2026	Each member	Submit an individual two-page checkpoint PDF with confirmed hours.
9-14 Nov 2026	Clara; Devano; Stacia	Integrate sensitivity and cluster-count results, finalise figures, and maintain the iteration log.
15-20 Nov 2026	All	Merge individual sections into the assessed report with one consistent argument and caveat set.
21-25 Nov 2026	All	Build slides, rehearse the story and run a final numerical consistency check.

Current blockers. The missing-output blocker is resolved and WORK_SPLIT.md now records the completed work. The remaining checkpoint items are human acceptance of cluster descriptions and confirmation of final course deadlines.

AI assistance and checks

AI assistance. I directed Codex to compare workflow versions, repair the notebook and draft this checkpoint. Codex produced the code and prose, while I reviewed the methodology and final evidence. The evidence checks include a clean notebook execution, 19 passing tests, archived-data loading and rendered report inspection. The cluster interpretation and final submission wording remain my responsibility.